# PILL BOARD

## NIRMATRELVIR BY PFIZER

**Suppliers in India**: N/A
**Drug Type**: Anti-viral
**Approval**: In process, decision awaited
**Included in MoHFW guidelines**: No
**Starting price**: N/A
**Efficacy**: Results from Phase II and III clinical trials showed it reduced risk of hospitalisation or death by 89 per cent

## MOLNUPIRAVIR BY MERCK

**Suppliers in India**: Cipla, Sun Pharma, Dr Reddy's Labs and nine other companies
**Drug type**: Anti-viral
**Approval**: Yes
**Included in MoHFW guidelines**: No
**Starting price**: Rs 1,399 for a 5-day course
**Efficacy**: 50 per cent reduction in hospitalisation and death in early-stage infections



## RONAPREVE BY ROCHE

**Suppliers in India**: Roche and Cipla
**Drug type**: Combination of two monoclonal antibodies, casirivimab and imdevimab"
**Approval**: Yes
**Included in MoHFW guidelines**: Yes
**Starting price**: Rs 59,750 per dose
**Efficacy**: Trials have shown it reduces hospital stays by four days and cut risk of death by a fifth

about a month and a half for cases in the UK to start declining for three days in a row, and roughly the same span for cases in the US to come down. In South Africa, the wave ended in approximately three weeks. Experts suggest that given the fact that around 80-85 per cent of the Omicron infections are asymptomatic and with the lightning fast speed at which the virus multiplies in a given population, the wave is expected to last for a shorter duration than previous ones.

In terms of lives and livelihoods too, Omicron is expected to be less disruptive than Delta. For one, the period of mandatory quarantine is far less than that for Delta. In December, the CDC halved the isolation time in the US from 10 days to five, and the UK government from 10 days to seven. In India, a person is assumed to be negative seven days after testing positive (or if there is no fever for three straight days) as opposed to 10 days in the last wave.

The Centre has also issued guidelines specifying that while the respective state governments must ensure adequate drugs, hospital beds and oxygen supply, they should impose location-specific curfews rather than complete lockdowns across the state or country. The thumb rule is that a city or town should be locked down only if there exist signs that its hospital infrastructure is likely to collapse given the sheer volume of cases.

## WILL OMICRON BESTOW NATURAL HERD IMMUNITY?

Herd immunity is the point at which a population acquires protection against a disease, either through vaccination or through naturally acquired immunity. Post the Delta wave, ICMR's fourth sero survey noted that almost 86 per cent of Indians had antibodies against the strain. And yet it was not enough to halt the third wave. Experts believe that while Omicron infection may protect a person against the strain in the future, there is no guarantee the immunity will stand up to future variants. "Nobody should deliberately try to get infected with Covid thinking that will give them immunity," says Dr Guleria. Experts fear that the natural vaccine theory, along with the belief that Omicron is mild, is breeding complacency among people towards Covid prevention measures. "Allowing unrestricted spread will also give the virus further chances to mutate," warns Dr Mishra.

Interestingly, experts have noted that the virus is developing mutations that will help it infect more people but not necessarily cause more severe disease, as with Omicron. "It is in the interest of the virus to survive. As we vaccinate and gain immunity against it, it will choose mutations that help it spread. This is a common process with most RNA respiratory viruses," says Dr Mishra. The Delta strain, for instance, had three mutations—L452R, P681R and E484Q—which, studies show, helped the virus replicate faster and evade antibodies more effectively, increasing its chances to trigger pneumonia and lung infection compared to the earlier Alpha variant. Omicron does not have these mutations. However, N501Y, a mutation that Omicron has in common with its Alpha, Beta, Gamma and Delta cousins, has been proven to increase transmissibility and facilitate immune escape. Omicron has three other mutations that are known to increase its transmissibility.

There is no guarantee, however, that Omicron's successor will be even less lethal. "Covid remains a concerning and unpredictable disease," warns virologist Dr Gagandeep Kang. "Even if Omicron is causing a milder disease, it does not mean Covid is over. We will see more waves and, in unvaccinated populations, we can also see the rise of new variants." Indeed, a new Covid variant, IHU, has been spotted in France. With 46 mutations, it is yet to be dubbed a variant of interest by the WHO. But it is a sure sign that Omicron is not the last we have seen of Covid-19. Constant vigilance is key if we are to avoid going through the entire Greek alphabet. ■